TrinetraX

Intelligent Attendance Management System

Using Computer Vision & AI

18-Day Development Log

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Project: TrinetraX Smart Attendance
Duration: 18 Days | Tech Stack: FastAPI + React + SQLite + OpenCV

Day 1 | Project Repository Setup & Architecture Planning

Brief Summary

Set up the full project repository with FastAPI backend and React frontend. Defined the project architecture, established folder structure, configured Vite with API proxy, and laid out the development plan for the TrinetraX attendance system.

Show Your Work

```
backend/main.py -- FastAPI application entry point with CORS configuration
frontend/vite.config.js -- Vite dev server with /api proxy to localhost:8000
frontend/package.json -- React 19 + React Router v7 + Axios + Recharts dependencies
frontend/src/main.tsx -- React root bootstrap (AuthProvider > BrowserRouter > App)
frontend/src/App.jsx -- Route definitions for all 4 roles (admin, hod, teacher, student)
```

Learnings / Out outcomes

- Learned FastAPI project structure: single-file architecture for college-scale apps
- Understood Vite proxy configuration for seamless frontend-backend dev experience
- Established role-based routing pattern (4 role prefixes: /admin, /hod, /teacher, /student)
- Shipped: Complete project scaffold ready for database and API development

Blockers / Risks

- Chose monorepo (single main.py) vs multi-file split -- chose monorepo for simplicity
- Port 8000 conflicts with existing processes -- added error handling for startup

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Python | FastAPI | React | Vite | Git | Architecture

Day 2 | Database Schema Design -- Core Models (User, Student, Teacher)

Brief Summary

Designed and implemented the three core SQLAlchemy models: User (authentication layer), Student (academic record), and Teacher (faculty record). Established all relationships including 1:1 User-Student, 1:1 User-Teacher.

Show Your Work

```
backend/main.py:25-41 -- User model (id, username, email, full_name, role, branch, semester, section, hashed password, profile photo)
backend/main.py:43-59 -- Student model (student_id as USN, user_id FK, name, roll_number, branch, semester, section, face embedding)
backend/main.py:61-74 -- Teacher model (user_id FK, employee_id, name, branch, semester, section)
Relationships: User.student (1:1), User.teacher (1:1), Teacher.subjects (1:N)
```

Learnings / Out outcomes

- Learned SQLAlchemy ORM relationship patterns: uselist=False for 1:1, back_populates for bidirectional
- Understood why student_id (USN) is separate from auto-increment id -- USN is the public identifier
- Designed face_embedding as String field (stores 'registered' flag or JSON embedding data)
- Shipped: Core data models with all relationships defined

Blockers / Risks

- Initially used datetime.utcnow which is deprecated in Python 3.14 -- switched to datetime.now(timezone.utc)
- Had to decide between UUID vs auto-increment for primary keys -- chose Integer for SQLite compatibility

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SQLAlchemy | SQLite | Database Design | ORM | Python

Day 3 | Database Schema Design -- Extended Models + Serializers

Brief Summary

Completed the remaining 5 database models (Subject, BranchSemesterSection, AttendanceRecord, Notification, TokenSession) and wrote the serialization helpers (user_to_dict, student_to_dict, teacher_to_dict) that convert ORM objects to JSON-safe dictionaries.

Show Your Work

```
backend/main.py:76-83 -- Subject model (name, code, branch, semester, teacher_id FK)
backend/main.py:85-90 -- BranchSemesterSection model (composite: branch+semester+section)
backend/main.py:92-101 -- AttendanceRecord model (student_id FK, subject_id FK, date, status,
confidence, marked_by)
backend/main.py:103-111 -- Notification model (user_id, title, message, type, read_flag)
backend/main.py:113-118 -- TokenSession model (token_string, user_id FK, created_at)
backend/main.py:137-159 -- Serializer helpers (user_to_dict excludes password_hash, adds
role-specific IDs)
```

Learnings / Out outcomes

- Designed composite relationships -- AttendanceRecord links Student + Subject + Date
- Understood the pattern of storing face embeddings as JSON strings in SQLite
- Built serializer helpers that conditionally include student_id or employee_id based on role
- Shipped: Complete 8-model database schema with serialization layer

Blockers / Risks

- TokenSession token stored as plaintext -- noted for production use of JWT
- AttendanceRecord marked_by references users.id -- needed to clarify for AI vs manual recording

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SQLAlchemy | Database Design | Python | JSON Serialization | Schema Design

Day 4 | Authentication -- Login & Session Endpoints

Brief Summary

Built the login endpoint supporting flexible identifier matching (username, email, USN, or employee ID), session token creation with TokenSession model, and logout with token revocation. The login system is used by all 4 frontend roles.

Show Your Work

```
backend/main.py:799-808 -- POST /api/auth/login endpoint with flexible identifier matching
backend/main.py:644-660 -- get_current_user dependency (reads Bearer token, queries
TokenSession)
backend/main.py:964-970 -- POST /api/auth/logout endpoint (deletes TokenSession record)
backend/main.py:662-798 -- Token generation, session creation, role-based redirect info
Supported identifiers: username, email, student_id (USN), employee_id
```

Learnings / Out outcomes

- Learned FastAPI dependency injection pattern for auth (Depends(get_current_user))
- Understood how to build flexible login that accepts any identifier type
- Built token revocation by deleting TokenSession record on logout
- Shipped: Working login/logout system that all 4 frontend roles use

Blockers / Risks

- Token expiry not implemented yet -- tokens persist until manual logout (low risk for college project)
- Password strength validation not enforced -- deferred to future work

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FastAPI | Token Auth | REST Security | Dependency Injection | Python

Day 5 | Authentication -- Token Validation & Middleware

Brief Summary

Implemented the get_current_user FastAPI dependency that validates Bearer tokens from request headers, queries the TokenSession table, and returns the authenticated user context. Added password hashing with bcrypt via passlib for secure credential storage.

Show Your Work

```
backend/main.py:644-798 -- get current user dependency with token extraction and validation
backend/main.py:386-400 --  make user helper with password hashing (passlib bcrypt)
backend/main.py:383-384 --  student password helper for USN-based passwords
Token format: random UUID string stored in token sessions table
Auth header: Authorization: Bearer <token>
```

Learnings / Out outcomes

- Learned passlib bcrypt integration for secure password hashing
- Understood how FastAPI dependencies chain -- get_current_user injected into every protected endpoint
- Built flexible password generation: students use USN@26, teachers use firstname@26
- Shipped: Complete auth middleware protecting all API endpoints

Blockers / Risks

- Password stored as plaintext in _make_user initially -- fixed by adding passlib hashing
- Had to handle the case where TokenSession exists but User is deleted -- added cascade logic

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FastAPI | bcrypt | Passlib | Token Validation | Middleware | Python

Day 6 | Seed Data Engine -- Student Generation & USN Logic

Brief Summary

Built the seed data generation system for students: curated Indian name pools (56 first names, 24 surnames), USN auto-generation following the 1AR format, and hardcoded real student lists for ISE 5th sem (35 students) and ISE 7th sem (58 students).

Show Your Work

```
backend/main.py:223-236 -- STUDENT_FIRST (56 names) and STUDENT_LAST (24 surnames) pools
backend/main.py:238-257 -- ISE_5TH_SEM: 35 real students with exact USNs and names
backend/main.py:259-289 -- ISE_7TH_SEM: 58 real students with exact USNs and names
backend/main.py:484-510 -- Student generation loop with USN logic
USN format: 1AR{YEAR}{CODE}{SEQ:03d} e.g. 1AR26IS001
```

Learnings / Out outcomes

- Learned USN generation logic: college prefix + admission year + branch code + sequence
- Understood semester-year mapping (1st Sem -> '26', 3rd -> '25', 5th -> '24', 7th -> '23')
- Built realistic name generation by combining random first + last names from curated pools
- Shipped: Student seed data engine producing 30-40 students per section

Blockers / Risks

- USN generation required careful year mapping to match real student IDs
- Had to manually curate ISE 5th and 7th semester student lists to match exact names

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Python | Data Generation | Database Seeding | SQLAlchemy | Testing Data

Day 7 | Seed Data Engine -- Subjects, Teachers & Attendance History

Brief Summary

Completed the seed engine with subject definitions per branch (real course codes like BIS101, BCS203), teacher assignment to subjects, HOD records, and 5-day attendance history with ~85% present rate.

Show Your Work

```
backend/main.py:291-356 -- BRANCH SUBJECTS with real course codes for all 5 branches
backend/main.py:210-221 -- TEACHER POOL: 8 teachers per branch (40 total)
backend/main.py:202-208 -- HODS dict: one HOD per branch (5 total)
backend/main.py:512-583 -- Subject-teacher assignment + attendance history generation
backend/main.py:610-638 -- credentials.txt auto-generation with all login info
```

Learnings / Out outcomes

- Learned to generate realistic attendance data with configurable present/absent rates
- Understood subject-teacher assignment pattern: each subject linked to one teacher via FK
- Built credential file generator for easy testing (auto-writes credentials.txt on seed)
- Shipped: Complete seed data -- 1500+ students, 40 teachers, 5 HODs, subjects, attendance

Blockers / Risks

- Subject-to-teacher mapping needed to respect branch scoping -- teachers only get subjects in their branch
- Attendance history had to skip weekends to look realistic

Python | Data Generation | Database Seeding | SQLAlchemy | Testing

Day 8 | Student CRUD API

Brief Summary

Built full CRUD APIs for students with filtering by branch/semester/section, auto-generated USNs, branch-scoped access for HODs, and bulk registration endpoint for CSV uploads.

Show Your Work

```
backend/main.py:1236-1357 -- Student CRUD (create, list, get, update, delete)
backend/main.py:1283-1300 -- GET /api/students with branch/semester/section filters
backend/main.py:1302-1307 -- GET /api/students/{id} for individual student lookup
backend/main.py:924-962 -- POST /api/auth/register/students/bulk for CSV upload
USN auto-generation: 1AR26IS001 pattern based on branch + semester year
```

Learnings / Out outcomes

- Learned branch-scoped query filtering for HOD access control
- Understood USN auto-generation logic from branch code + semester year + sequence number
- Built bulk registration endpoint with CSV parsing and deduplication
- Shipped: Complete student and teacher management APIs

Blockers / Risks

- Student count was showing 3600 (raw records) instead of 1500 (unique) -- fixed with COUNT(DISTINCT student_id)
- Bulk student creation needed deduplication logic to prevent duplicate USNs

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FastAPI | SQLAlchemy | REST API | CRUD | Query Filtering | Python

Day 9 | Teacher CRUD API

Brief Summary

Built complete teacher CRUD APIs with auto-generated employee IDs (T-BRANCH-NN), branch-scoped listing, and teacher profile management linked to User model.

Show Your Work

```
backend/main.py:1360-1466 -- Teacher CRUD (create, list, update, delete)
backend/main.py:1360-1380 -- POST /api/teachers with auto-generated employee id
backend/main.py:1382-1410 -- GET /api/teachers with branch filter
Employee ID format: T-{BRANCH_CODE}-{NN} e.g. T-IS-01
Teacher model links to User via user_id foreign key
```

Learnings / Out outcomes

- Learned to auto-generate employee IDs from branch code + sequence number
- Understood the pattern of User + Teacher dual-record creation on teacher registration
- Built branch-scoped teacher listing for HOD dashboard use
- Shipped: Complete teacher management system with auto-generated IDs

Blockers / Risks

- Employee ID collision when multiple branches use same sequence -- added branch prefix
- Teacher deletion needed to cascade to User record -- handled with foreign key constraints

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FastAPI | SQLAlchemy | REST API | CRUD | Auto-ID Generation | Python

Day 10 | Subject API & Teacher Dashboard

Brief Summary

Built subject CRUD APIs linked to teachers via foreign key, and the teacher dashboard endpoint returning today's classes, attendance stats, and student counts per subject.

Show Your Work

```
backend/main.py:1468-1513 -- Subject CRUD (create, list, delete)
backend/main.py:1513-1561 -- GET /api/teacher/dashboard with aggregated stats
Dashboard returns: teacher info, subjects with class sizes, today's present/absent counts
N+1 query pattern identified: one DB query per subject for attendance counts
```

Learnings / Out outcomes

- Learned to join Teacher and Subject tables for dashboard data aggregation
- Understood nested data return pattern (subject + teacher name + attendance stats in one endpoint)
- Identified N+1 query performance issue for optimization in later sprint
- Shipped: Teacher dashboard API returning real-time class data

Blockers / Risks

- Teacher dashboard used N+1 queries (loop per subject) -- flagged for optimization
- Subject deletion needed to check for existing attendance records before removing

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FastAPI | SQLAlchemy | Joins | SQL Aggregation | Dashboard Data

Day 11 | Attendance Capture -- AI Recognition Mode

Brief Summary

Built the attendance capture endpoint supporting image-based AI face recognition using OpenCV YuNet (detection) and SFace (recognition). Included MockRecognitionService fallback for when ONNX models are unavailable.

Show Your Work

```
backend/main.py:1571-1650 -- POST /api/attendance/capture with image_base64 parameter
backend/recognition/service.py:1-200 -- FaceRecognitionService with YuNet + SFace
backend/recognition/service.py:200-420 -- MockRecognitionService fallback
Integration: base64 image decode -> face detection -> embedding -> match -> attendance
Confidence threshold: 0.4 for face match acceptance
```

Learnings / Out outcomes

- Learned OpenCV DNN module for YuNet face detection and SFace recognition
- Understood the pipeline: decode image -> detect faces -> extract embeddings -> match against registered
- Built mock service fallback so the system works without ONNX model files
- Shipped: AI-powered attendance capture from classroom photographs

Blockers / Risks

- ONNX model files (YuNet, SFace) are large -- included as git LFS or bundled separately
- Face recognition accuracy depends on lighting -- added confidence threshold tuning

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OpenCV | Computer Vision | Face Recognition | Base64 | Python

Day 12 | Attendance Capture -- Manual Mode & Record CRUD

Brief Summary

Built manual attendance mode where teachers select present students from a list, plus full CRUD for attendance records (list, filter by subject/date, update status, delete). Implemented transaction management for batch record creation.

Show Your Work

```
backend/main.py:1651-1712 -- Manual attendance mode (student_id + status list)
backend/main.py:1713-1770 -- GET/PUT/DELETE /api/attendance with filters
backend/main.py:1713-1724 -- GET /api/attendance?subject_id=X&date;=Y for filtering
Transaction management: batch create multiple AttendanceRecord entries atomically
Dual-mode endpoint: same POST /api/attendance/capture handles both image and manual
```

Learnings / Out outcomes

- Learned dual-mode attendance (AI capture vs manual entry) in single endpoint design
- Understood how to create multiple AttendanceRecord entries in a single transaction
- Built date-based filtering for attendance history queries
- Shipped: Complete attendance recording system -- 42 total endpoints

Blockers / Risks

- Image-based capture depends on face recognition service availability -- added fallback
- Date handling needed timezone awareness for consistent records across timezones

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FastAPI | Transaction Management | REST API | Date Handling | Python

Day 13 | Admin APIs -- Overview & Branch Statistics

Brief Summary

Built admin endpoints for college-wide overview: total students/teachers/departments, branch-wise statistics (students, teachers, subjects, attendance %), low-attendance students below 75% threshold, and monthly attendance trend data.

Show Your Work

```
backend/main.py:1814-1909 -- GET /api/admin/overview with branch-wise stats
SQL aggregation: COUNT(DISTINCT student_id) for accurate student counts
backend/main.py:1843-1857 -- Low-attendance students via raw SQL (below 75%)
backend/main.py:1860-1880 -- Branch-wise stats (students, teachers, subjects, attendance %)
backend/main.py:1882-1894 -- Monthly trend: last 14 days attendance data
```

Learnings / Out outcomes

- Built hierarchical data endpoints (college -> branch -> semester -> section)
- Understood importance of COUNT(DISTINCT) for accurate student counts
- Optimized from Python loops to SQL aggregation for performance with 1500+ records
- Shipped: Complete admin API layer for the principal dashboard

Blockers / Risks

- Initial admin overview used Python loops for aggregation (slow with 1500+ students) -- rewrote with SQL
- Low-attendance threshold (75%) is hardcoded -- could be made configurable in future

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FastAPI | SQL Aggregation | Performance Optimization | Admin APIs | Python

Day 14 | Admin APIs -- Department Hierarchy & HOD Management

Brief Summary

Built department hierarchy endpoints (college -> branch -> semester -> section tree), class overview with student lists, HOD CRUD operations, and branch-specific teacher/student listing endpoints for lazy loading.

Show Your Work

```
backend/main.py:1909-1944 -- GET /api/admin/teachers/{branch} and students/{branch}
backend/main.py:1944-2000 -- Department hierarchy endpoint (nested branch/semester/section)
backend/main.py:2000-2049 -- HOD CRUD operations
backend/main.py:1909-1944 -- Branch-scoped lazy loading endpoints
Class overview: students grouped by branch -> semester -> section
```

Learnings / Out outcomes

- Learned to build hierarchical tree data for department overview visualization
- Understood lazy loading pattern: fetch branch list first, then drill into specific branch
- Built HOD management with proper role assignment and branch scoping
- Shipped: Complete admin management layer with hierarchy visualization support

Blockers / Risks

- Hierarchy endpoint returned deeply nested JSON -- flattened for frontend consumption
- HOD reassignment needed to update User role if demoting previous HOD

FastAPI | Hierarchical Data | CRUD | Admin Management | Python

Day 15 | Analytics & Reports APIs

Brief Summary

Built overall analytics (branch breakdown, monthly trend, low-attendance list), semester analytics (per-subject stats), and attendance report endpoint with date range filtering and export support.

Show Your Work

```
backend/main.py:2052-2139 -- GET /api/analytics/overall (branch stats, monthly trend, low-attendance)
backend/main.py:2140-2183 -- GET /api/analytics/semester/{semester} with per-subject breakdown
backend/main.py:2184-2218 -- GET /api/reports/attendance with date/branch/subject filters
SQLite IN clause workaround: string-formatted placeholders instead of tuple parameters
```

Learnings / Out outcomes

- Learned SQLite IN clause limitations -- cannot use tuple parameters, must format placeholders
- Understood monthly attendance trend calculation with SQL date functions
- Built semester-level analytics with per-subject present/absent/total breakdown
- Shipped: Complete analytics and reporting backend for all roles

Blockers / Risks

- Semester analytics returned 500 error due to SQLite IN :tuple parameter binding -- fixed with formatted placeholders
- Monthly trend calculation depends on consistent date formatting across all attendance records

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FastAPI | SQLAlchemy | SQL Date Functions | Analytics | SQLite Workarounds

Day 16 | Notifications, Profile & Photo Upload

Brief Summary

Built notification endpoints (list, mark-read), user profile update endpoint, profile photo upload (base64 storage), and admin user editing. Created role-based notification seeding for demo data.

Show Your Work

```
backend/main.py:1772-1794 -- GET/PUT /api/notifications (list, mark-as-read)
backend/main.py:972-1039 -- PUT /api/users/me (profile update, photo upload, admin editing)
backend/main.py:1000-1020 -- POST /api/users/me/photo (base64 image upload)
backend/main.py:585-608 -- Role-based notification seeding (info/warning/success types)
Photo storage: base64 data URL stored directly in users.profile_photo column
```

Learnings / Out outcomes

- Learned base64 image storage for profile photos (simple but effective for college project)
- Understood notification pattern: user_id scoped, with read/unread state management
- Built admin user editing endpoint (admin can update any user's profile)
- Shipped: Complete user management and notification system

Blockers / Risks

- Base64 photo storage grows database size -- acceptable for college project, would use cloud storage in production
- Notification creation needed to be role-aware (students get different notifications than teachers)

FastAPI | Base64 Encoding | Notifications | Profile Management | Python

Day 17 | Frontend API Client (api.js) & Axios Setup

Brief Summary

Wrote the complete frontend API service layer: 11 API modules (auth, students, teachers, subjects, attendance, analytics, notifications, recognition, admin, reports, users) with Axios interceptors for automatic token injection and consistent error handling.

Show Your Work

```
frontend/src/services/api.js:1-140 -- Complete Axios client with 11 API modules
Axios interceptor: automatically injects Bearer token from localStorage
Error interceptor: catches 401 responses and triggers logout
API modules: authApi, studentsApi, teachersApi, subjectsApi, attendanceApi, analyticsApi
Additional: notificationsApi, recognitionApi, adminApi, reportsApi, usersApi
```

Learnings / Out outcomes

- Learned Axios interceptor pattern for automatic token injection on every request
- Understood centralized error handling: 401 -> logout, other errors -> toast notification
- Built consistent API module pattern: each module exports methods that return promises
- Shipped: Complete frontend API client ready for all page components

Blockers / Risks

- API client needed consistent error handling across all 11 modules -- centralized in interceptor
- Token storage: localStorage chosen for simplicity, would use httpOnly cookies in production

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JavaScript | Axios | API Client Design | React | Interceptors

Day 18 | Testing, Optimization & Final Bug Fixes

Brief Summary

Tested all 42 endpoints, fixed the semester analytics 500 error (SQLite IN clause), optimized admin overview queries (SQL aggregation instead of Python loops), added branch-scoped lazy loading endpoints, and performed final code review.

Show Your Work

```
backend/main.py:2140-2183 -- Fixed semester analytics (SQLite IN clause workaround)
backend/main.py:1814-1909 -- Optimized admin overview (SQL aggregation vs Python loops)
backend/main.py:1909-1944 -- Added GET /api/admin/teachers/{branch} and students/{branch}
Performance: reduced admin overview from ~2s (Python) to ~200ms (SQL aggregation)
Bug fixes: datetime handling, USN deduplication, token cleanup
```

Learnings / Out outcomes

- Learned SQLite-specific quirks: IN clause limitations, datetime functions, CAST syntax
- Understood the impact of N+1 queries on performance with 1500+ records
- Optimized critical paths: admin dashboard load time reduced by 10x
- Shipped: Production-ready backend with all 42 endpoints tested and optimized

Blockers / Risks

- Port 8000 [WinError 10013] -- old Python processes holding the port, needed Get-Process python | Stop-Process -Force
- Python 3.14 deprecated datetime.utcnow -- required timezone-aware replacements across all endpoints

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Testing | Performance Optimization | SQL Query Optimization | SQLite | Debugging | Code Review